

BRATSK, Russian Federation

WMO: 303090

Lat: 56.371N

Lon: 101.698E

Elev: 491

StdP: 95.57

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-34.0	-40.3	0.1	-37.3	-37.0	0.1	-33.9	9.7	-14.5	8.3	-15.3	1.6	290	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.2	18.4	26.4	17.9	24.9	17.1	20.0	25.7	19.0	24.5	18.1	23.3	2.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.8	22.8	17.0	12.9	21.7	16.0	12.1	20.6	59.8	25.9	56.3	24.5	53.1	23.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.7	6.8	-38.0	31.3	5.0	1.1	-41.6	32.1	-44.5	32.7	-47.3	33.4	-50.9	34.2		
(5)			-38.1	21.9	5.0	1.1	-41.6	22.7	-44.5	23.3	-47.3	23.9	-50.9	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-20.1	-16.1	-7.8	0.8	8.3	15.8	18.5	15.5	7.9	-0.3	-10.9	-17.8		
	DBStd	14.56	7.98	7.15	6.40	4.89	5.01	4.35	3.18	3.78	4.44	5.38	7.89	7.80		
	HDD10.0	4504	934	732	552	277	95	6	0	5	93	321	627	863		
	HDD18.3	6936	1192	965	810	526	313	98	38	102	315	578	877	1121		
	CDD10.0	691	0	0	0	1	42	180	263	175	29	1	0	0		
	CDD18.3	82	0	0	0	0	2	22	43	14	1	0	0	0		
	CDH23.3	687	0	0	0	0	24	225	316	119	4	0	0	0		
	CDH26.7	135	0	0	0	0	3	47	67	18	0	0	0	0		
	WSAvg	3.0	2.5	2.6	3.2	3.7	3.6	2.8	2.5	2.6	3.0	3.3	3.2	2.6		
(15) Precipitation	PrecAvg	369	17	13	12	17	34	50	56	60	38	25	26	20		
	PrecMax	485	37	25	26	42	71	144	157	105	89	39	39	38		
	PrecMin	247	7	5	4	3	7	3	10	10	2	11	11	11		
	PrecStd	56	7	5	6	9	15	28	35	24	19	9	7	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.1	0.0	7.6	16.3	26.1	30.1	30.9	28.9	23.2	14.0	6.8	-0.7		
		MCWB	-3.0	-2.6	2.4	7.7	14.9	18.2	20.5	19.2	14.8	8.3	3.9	-2.4		
	2%	DB	-4.9	-2.3	5.0	13.3	23.0	28.0	28.6	26.2	19.9	11.1	3.2	-3.8		
		MCWB	-5.6	-4.0	0.6	6.2	12.7	17.3	19.4	18.2	13.7	6.9	1.1	-4.5		
	5%	DB	-7.2	-4.1	2.9	10.9	20.0	26.0	26.8	24.3	17.1	9.0	1.2	-5.7		
		MCWB	-7.9	-5.3	-0.5	4.9	11.0	16.8	18.7	17.3	12.0	5.7	-0.4	-6.4		
	10%	DB	-9.9	-6.2	0.9	8.1	17.1	24.0	25.0	22.3	14.9	6.9	-0.4	-7.5		
		MCWB	-10.3	-7.4	-1.9	3.5	9.6	15.9	18.0	16.4	10.8	4.4	-1.5	-8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	-2.2	2.7	9.0	15.6	20.5	22.0	20.2	16.4	9.3	3.9	-2.0		
		MCDB	-2.6	-0.8	6.6	14.0	24.2	26.6	28.1	26.0	20.8	12.4	5.8	-0.8		
	2%	WB	-5.5	-3.8	1.0	7.1	13.5	18.9	20.6	19.1	14.2	7.6	1.3	-4.4		
		MCDB	-4.8	-2.6	4.3	12.1	21.5	25.3	26.2	24.8	18.7	10.2	2.8	-3.7		
	5%	WB	-8.1	-5.4	-0.5	5.2	11.8	17.7	19.6	18.0	12.7	6.1	-0.2	-6.3		
		MCDB	-7.5	-4.2	2.5	10.0	18.2	24.0	24.9	23.3	16.6	8.7	1.1	-5.7		
	10%	WB	-10.4	-7.5	-1.8	3.6	10.2	16.6	18.6	16.9	11.3	4.3	-1.7	-8.2		
		MCDB	-10.1	-6.5	0.7	7.9	16.3	22.3	23.6	21.5	14.3	6.4	-0.5	-7.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.5	9.8	9.2	10.7	10.8	10.0	9.8	8.8	7.2	7.1	6.7		
		MCDBR	7.5	8.5	10.6	12.2	14.6	13.8	12.4	12.6	12.1	10.2	6.5	7.1		
	5% WB	MCWBR	7.2	7.3	7.7	7.6	7.6	6.4	6.1	6.4	7.0	6.7	5.6	6.7		
		MCWBR	7.2	8.5	9.8	11.2	13.4	11.9	11.1	11.6	11.4	9.2	6.6	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.300	0.326	0.359	0.395	0.416	0.434	0.392	0.342	0.329	0.287	0.264		
	taud		2.018	2.029	2.053	2.216	2.237	2.257	2.225	2.326	2.432	2.286	2.090	1.977		
	Ebn at Noon		585	731	817	854	843	827	800	815	813	707	563	479		
	Edh at Noon		64	96	121	121	127	126	128	108	84	75	59	51		
(44) All-Sky Solar Radiation	RadAvg		0.79	1.77	3.23	4.56	5.21	5.78	5.32	4.33	2.79	1.48	0.75	0.49		
	RadStd		0.07	0.10	0.19	0.31	0.47	0.42	0.46	0.27	0.33	0.15	0.09	0.07		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page